

One workload, one contract, two optimizers

Evidence contract

ON qa_service ASSERT

$P(\text{incorrect}) \leq 0.10$

$P(\text{uncited claim}) \leq 0.10$

$P(\text{evidence} > 72\text{h}) \leq 0.10$

$P(\text{latency} > 10\text{s}) \leq 0.05$

WITH CONFIDENCE 0.9

MINIMIZE llm_cost

Rewrite lattice $\rightarrow$ policies

P3 hybrid+rerank+NLI+KG · max **7.2 mCNY**

P2 hybrid+rerank · plus **1.7 mCNY**

P1 hybrid retrieval · flash **0.23 mCNY**

P0 sparse-only · flash **0.17 mCNY**

costs span 42×

Point-estimate optimizer

picks cheapest plan whose measured risk clears α on validation data

true violation $> \alpha$ in 37-64% of draws — and it never knows

ContractRAG (this paper)

certifies by fixed-sequence risk tests;
deploys cheapest certified $\hat{\pi}$;
e-process monitor guards drift

**$\Pr(\text{constraints hold}) \geq 0.9$
worst case 7%, up to 20× cheaper**